

Documentation

STEP 1

Code:

```
import pandas as pd  
import numpy as np
```

```
import matplotlib.pyplot as plt  
import seaborn as sns
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 2

Markdown Content:

NumPy for numerical computations ,Pandas was used for data manipulation, Matplotlib for creating graphical representations, and Seaborn for generating advanced statistical visualizations. These libraries provide the foundation for preprocessing, analyzing, and interpreting the dataset effectively.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 3

Markdown Content:

pandas is used to load and manipulate the dataset.
numpy is used for mathematical calculations.
matplotlib is used to create graphs and visualizations.
seaborn is used to create attractive statistical plots.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 4

Code:

```
df = pd.read_csv("/content/RealEstate_HousePrice_Dataset_4200 -  
RealEstate_HousePrice_Dataset_4200.csv.csv")
```

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STEP 5

Markdown Content:

The dataset was loaded into a Pandas DataFrame to enable data analysis, preprocessing, and visualization.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 6

Code:

```
df.head()
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

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STEP 7

Markdown Content:

This code displays the first 5 rows of the dataset to understand its structure and columns.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 8

Code:

df.tail()

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 9

Markdown Content:

This code displays the Last 5 rows of the dataset to understand its structure and columns.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 10

Code:

df.shape

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

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STEP 11

Markdown Content:

This code Shows the number of rows and columns present in the dataset.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 12

Code:

df.info()

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

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STEP 13

Markdown Content:

Provides information about:

Data types

Number of non-null values

Memory usage

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 14

Code:

```
df.duplicated().sum()
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 15

Markdown Content:

This code provide total sum of duplicate value present in the Dataset.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 16

Code:

```
df.isnull().sum()
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 17

Markdown Content:

This code provide total sum of null values present in the Dataset.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 18

Code:

```
(df.isnull().sum() / len(df)) * 100
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 19

Markdown Content:

This code provide total sum of null values in % in the dataset

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 20

Code:

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
plt.figure(figsize=(12,6))
```

```
sns.heatmap(
    df.isnull(),
```

```
cbar=False,  
cmap='viridis'  
)
```

```
plt.title("Missing Values Heatmap")
```

```
plt.show()
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 21

Markdown Content:

This code used for checking any null values present in the dataset through visulazation.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 22

Code:

```
plt.figure(figsize=(12,8))
```

```
sns.heatmap(  
    df.corr(numeric_only=True),  
    annot=True,  
    cmap='coolwarm'  
)
```

```
plt.title("Correlation Heatmap")
```

```
plt.show()
```

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STEP 23

Markdown Content:

This graph shows the strength of relationships between variables.

Values close to +1 indicate strong positive correlation.

Values close to -1 indicate strong negative correlation.

Values close to 0 indicate weak correlation.

This helps identify which features influence house prices.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 24

Markdown Content:

Shows relationships among numerical variables and helps identify features strongly associated with house price.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 25

Code:

```
plt.figure(figsize=(8,5))
```

```
sns.scatterplot(  
    x='area_sqft',  
    y='house_price_inr',  
    data=df,  
    hue='house_price_inr',  
    palette='viridis'  
)
```

```
plt.title('Area vs House Price')
```

```
plt.show()
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

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STEP 26

Markdown Content:

This plot shows the relationship between house area and house price.

If points move upward from left to right, it indicates that larger houses generally have higher prices.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 27

Markdown Content:

Visualizes the relationship between house area and house price. Larger properties generally tend to have higher prices.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 28

Code:

```
plt.figure(figsize=(8,5))
```

```
sns.scatterplot(  
    x='location_score',  
    y='house_price_inr',  
    data=df,  
    color='green'  
)
```

```
plt.title('Location Score vs House Price')  
plt.show()
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 29

Markdown Content:

This plot shows the relationship between Location score and house price.

If points move upward from left to right, it shows that larger houses have higher prices.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 30

Markdown Content:

Examines how location quality affects house prices. Better locations usually result in higher property values.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 31

Code:

```
plt.figure(figsize=(8,5))
```

```
sns.scatterplot(  
    x='age_years',  
    y='house_price_inr',  
    data=df  
)
```

```
plt.title('Age vs House Price')  
plt.show()
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 32

Markdown Content:

This plot shows the relationship between age and house price.

If points move upward from left to right, it shows that larger houses have higher prices.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 33

Code:

```
plt.figure(figsize=(8,5))

sns.histplot(
    df['house_price_inr'],
    bins=30,
    kde=True
)

plt.title("Distribution of House Prices")
plt.xlabel("House Price")
plt.ylabel("Frequency")

plt.show()
```

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Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 34

Markdown Content:

Shows how house prices are distributed across the dataset and helps identify skewness.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 35

Code:

```
plt.figure(figsize=(8,5))

sns.histplot(
    df['area_sqft'],
    bins=30,
    kde=True
)
```

```
)
```

```
plt.title("Distribution of House Area")
```

```
plt.show()
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 36

Markdown Content:

Displays the distribution of house sizes and helps understand the range of property areas.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 37

Code:

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
plt.figure(figsize=(8,5))
```

```
sns.countplot(
    x='bedrooms',
    data=df,
    hue='bedrooms',
    palette='Set2',
    legend=False
)
```

```
plt.title("Distribution of Bedrooms")
```

```
plt.show()
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 38

Markdown Content:

This code used to visulized the frequency of houses with different numbers of bedrooms.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 39

Code:

```
cols = [  
    'area_sqft',  
    'age_years',  
    'distance_city_km',  
    'lot_size_sqft',  
    'renovation_years_ago',  
    'house_price_inr'  
]  
  
df_clean = df.copy()  
  
for col in cols:  
  
    Q1 = df_clean[col].quantile(0.25)  
    Q3 = df_clean[col].quantile(0.75)  
  
    IQR = Q3 - Q1  
  
    lower = Q1 - 1.5 * IQR  
    upper = Q3 + 1.5 * IQR  
  
    df_clean[col] = np.where(  
        df_clean[col] < lower,  
        lower,  
        df_clean[col]  
    )  
  
    df_clean[col] = np.where(  
        df_clean[col] > upper,  
        upper,  
        df_clean[col]  
    )
```

```
df_clean[col] > upper,  
upper,  
df_clean[col]  
)
```

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STEP 40

Markdown Content:

This code detects outliers in six numerical features using the IQR method. For each feature, the first quartile (Q1), third quartile (Q3), and Interquartile Range (IQR) are calculated. Lower and upper limits are then determined using the formula $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$. Instead of deleting outliers, extreme values are replaced with the nearest boundary values. This process reduces the impact of outliers while preserving all records in the dataset, making the data more suitable for machine learning model training.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 41

Code:

```
import matplotlib.pyplot as plt  
import seaborn as sns  
  
numerical_cols = [  
    'area_sqft',  
    'age_years',  
    'distance_city_km',  
    'lot_size_sqft',  
    'renovation_years_ago',  
    'house_price_inr'  
]  
  
colors = ['red', 'blue', 'green', 'orange', 'purple', 'brown']  
  
for col, color in zip(numerical_cols, colors):
```

```
plt.figure(figsize=(8,4))
```

```
sns.boxplot(  
    x=df[col],  
    color=color  
)
```

```
plt.title(f'Boxplot of {col}')  
plt.show()
```

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STEP 42

Markdown Content:

This code is used to a Boxplots help identify outliers observations in dataset.

Points outside the whiskers may represent extremely high or low property prices.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 43

Code:

```
numerical_cols = [  
    'area_sqft',  
    'age_years',  
    'distance_city_km',  
    'lot_size_sqft',  
    'renovation_years_ago',  
    'house_price_inr'  
]
```

```
for col in numerical_cols:  
    Q1 = df[col].quantile(0.25)  
    Q3 = df[col].quantile(0.75)  
    IQR = Q3 - Q1
```

```
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
```

```
outliers = df[(df[col] < lower_bound) | (df[col] > upper_bound)]
```

```
print(f'{col}: {len(outliers)} outliers')
```

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Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 44

Markdown Content:

This code identifies the number of outliers present in each numerical feature using the Interquartile Range (IQR) method. For every numerical column, the first quartile (Q1), third quartile (Q3), and IQR are calculated. Based on these values, lower and upper boundaries are determined using the formulas $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$. Any data point falling outside these boundaries is considered an outlier. The code then counts and displays the total number of outliers for each feature. This analysis helps understand the extent of extreme values in the dataset before applying any outlier treatment or machine learning techniques.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 45

Code:

```
X = df_clean[
    [
        'area_sqft',
        'bedrooms',
        'bathrooms',
        'location_score',
        'age_years'
    ]
]

y = df_clean['house_price_inr']
```

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STEP 46

Markdown Content:

This step performs feature selection and target variable definition for the machine learning model. Five important housing attributes—area, number of bedrooms, number of bathrooms, location score, and house age—are selected as input features and stored in X. The house price is selected as the target variable and stored in y. These variables will be used in subsequent steps for model training, testing, and prediction.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 47

Code:

```
print(X.shape)
print(y.shape)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 48

Markdown Content:

This code is used to verify the dimensions of the feature matrix (X) and target variable (y). It ensures that the number of records in both datasets is the same and confirms that the data has been prepared correctly for machine learning model training. The feature matrix contains the selected input attributes, while the target variable contains the corresponding house prices.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 49

Code:

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 50

Markdown Content:

This step divides the dataset into training and testing subsets. The training data, consisting of 80% of the records, is used to train the machine learning model, while the remaining 20% is used to evaluate its performance on unseen data. The random state is fixed to ensure consistent and reproducible results across multiple executions.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 51

Code:

```
print("Training Features :", X_train.shape)
print("Testing Features :", X_test.shape)

print("Training Target :", y_train.shape)
print("Testing Target :", y_test.shape)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 52

Markdown Content:

This code verifies the dimensions of the training and testing datasets after splitting. It confirms that the feature and target variables have been divided correctly and that the number of records in the training and testing sets matches the specified split ratio. This validation step ensures that the data is properly prepared for machine learning model training and evaluation.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 53

Code:

```
X.hist(  
    figsize=(12,8),  
    bins=20  
)
```

```
plt.tight_layout()  
plt.show()
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 54

Markdown Content:

This code visualizes the distribution of all selected input features using histograms. Each histogram shows how frequently different values occur within a feature. The visualization helps identify patterns such as normal distribution, skewness, spread of data, and potential outliers. Understanding feature distributions is an important step in exploratory data analysis (EDA) before building a machine learning model.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 55

Code:

```
X_simple = df_clean[['area_sqft']]
```

```
y_simple = df_clean['house_price_inr']
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 56

Markdown Content:

This step prepares the dataset for Simple Linear Regression. Only one feature, `area_sqft`, is selected as the independent variable because house area is often one of the most influential factors affecting property prices. The target variable is `house_price_inr`, which represents the house price to be predicted. The relationship between house area and house price will be analyzed using a simple linear regression model.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 57

Markdown Content:

Simple Linear Regression

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 58

Code:

```
X_train_s, X_test_s, y_train_s, y_test_s = train_test_split(  
    X_simple,  
    y_simple,  
    test_size=0.20,  
    random_state=42  
)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 59

Markdown Content:

This step divides the simple regression dataset into training and testing subsets. The training set contains 80% of the observations and is used to train the Simple Linear Regression model. The testing set contains the remaining 20% of observations and is used to evaluate the model's predictive performance on unseen data. Using a fixed random state ensures consistent and reproducible results.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 60

Code:

```
from sklearn.linear_model import LinearRegression
```

```
slr = LinearRegression()
```

```
slr.fit(X_train_s, y_train_s)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 61

Markdown Content:

This step creates and trains a Simple Linear Regression model using house area as the independent variable and house price as the dependent variable. The model learns the relationship between these variables from the training data and generates a mathematical equation that can be used to predict house prices based on the area of a house.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 62

Code:

```
pred_simple = slr.predict(X_test_s)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 63

Markdown Content:

This step generates house price predictions using the trained Simple Linear Regression model. The model applies the relationship learned during training to the unseen testing data and estimates the house prices based on house area. These predicted values will later be compared with the actual house prices to evaluate the model's accuracy and performance.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 64

Markdown Content:

House Price= $m \times \text{Area} + c$

m = Slope (coefficient)

c = Intercept

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 65

Code:

```
print("Slope:", slr.coef_[0])
```

```
print("Intercept:", slr.intercept_)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 66

Markdown Content:

This code retrieves the parameters of the trained Simple Linear Regression model. The slope represents the change in house price for each additional square foot of area, while the intercept represents the baseline predicted price when the area is zero. Together, these values form the regression equation used for house price prediction.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 67

Markdown Content:

House Price=(Slope×Area)+Intercept

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 68

Markdown Content:

Price=(4850.75×1000)+250000.50

Price=5,100,750.50

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 69

Code:

plt.figure(figsize=(8,5))

plt.scatter(

X_test_s,

y_test_s,

```
alpha=0.5
)

plt.plot(
    X_test_s,
    pred_simple,
    color='red'
)

plt.xlabel("Area (sqft)")
plt.ylabel("House Price")
plt.title("Simple Linear Regression")

plt.show()
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 70

Markdown Content:

This code is used to visualize the Best Fit Line of the Simple Linear Regression model.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 71

Markdown Content:

This code visualizes the results of the Simple Linear Regression model. The scatter plot represents the actual house prices from the testing dataset, while the red line represents the predicted prices generated by the regression model. The graph helps evaluate how well the regression line fits the actual data and illustrates the relationship between house area and house price.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 72

Code:

```

from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

mse_simple = mean_squared_error(
    y_test_s,
    pred_simple
)

mae_simple = mean_absolute_error(
    y_test_s,
    pred_simple
)

rmse_simple = np.sqrt(mse_simple)

r2_simple = r2_score(
    y_test_s,
    pred_simple
)

print("MSE :", mse_simple)
print("MAE :", mae_simple)
print("RMSE :", rmse_simple)
print("R2 Score :", r2_simple)

```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 73

Markdown Content:

This code evaluates the performance of the Simple Linear Regression model using four commonly used regression metrics: Mean Squared Error (MSE), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. These metrics measure the difference between actual and predicted house prices and indicate how accurately the model performs on unseen test data. Lower error values and a higher R^2 Score indicate better model performance.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 74

Markdown Content:

MSE stands for Mean Squared Error, MAE stands for Mean Absolute Error, RMSE stands for Root Mean Squared Error, and R^2 Score stands for Coefficient of Determination. These metrics are used to evaluate the performance and accuracy of a Linear Regression model.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 75

Code:

```
n = len(y_test_s)

p = 1

adj_r2_simple = 1 - (
    ((1-r2_simple)*(n-1))
    /
    (n-p-1)
)

print("Adjusted R2 :", adj_r2_simple)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 76

Markdown Content:

This code calculates the Adjusted R^2 Score of the Simple Linear Regression model. Adjusted R^2 measures how well the independent variable explains the variation in the dependent variable while accounting for the number of predictors used in the model. It provides a more reliable evaluation metric than R^2 when comparing regression models with different numbers of features.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 77

Code:

```
residuals = y_test_s - pred_simple

plt.figure(figsize=(8,5))

sns.scatterplot(
    x=pred_simple,
    y=residuals
)

plt.axhline(
    y=0,
    color='red'
)

plt.xlabel("Predicted Values")
plt.ylabel("Residuals")

plt.title("Residual Plot")

plt.show()
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 78

Markdown Content:

This code creates a Residual Plot to analyze the prediction errors of the Linear Regression model. The residuals are calculated as the difference between actual and predicted house prices. If the points are randomly scattered around the zero line, the model is considered a good fit for the data.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 79

Code:

```
plt.figure(figsize=(8,5))
```

```
sns.histplot(  
    residuals,  
    kde=True  
)
```

```
plt.title("Residual Distribution")
```

```
plt.show()
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 80

Markdown Content:

This code creates a Residual show distribution.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 81

Markdown Content:

Residual=Actual Value–Predicted Value

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 82

Code:

```
X = df_clean[  
    [  
        'area_sqft',  
        'bedrooms',  
        'bathrooms',  
        'location_score',  
        'age_years'
```

```
y = df_clean['house_price_inr']
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 83

Markdown Content:

This code prepares the dataset for Multiple Linear Regression. It selects five important house features (area, bedrooms, bathrooms, location score, and age) as input variables and house price as the output variable. The model will learn how these factors affect house prices and use them to make predictions.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 84

Code:

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 85

Markdown Content:

This code divides the dataset into training and testing sets. The training data is used to teach the Multiple Linear Regression model, while the testing data is used to evaluate its performance on unseen data.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 86

Code:

```
mlr = LinearRegression()
```

```
mlr.fit(X_train, y_train)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 87

Code:

```
pred_multi = mlr.predict(X_test)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 88

Markdown Content:

This code uses the trained Multiple Linear Regression model to predict house prices for the test dataset.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 89

Code:

```
coefficients = pd.DataFrame({  
    'Feature': X.columns,  
    'Coefficient': mlr.coef_  
})
```

coefficients

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 90

Markdown Content:

This code displays the contribution of each feature in the Multiple Linear Regression model. The coefficient values indicate the direction and magnitude of the relationship between each feature and house price. Positive coefficients increase the predicted price, while negative coefficients decrease it.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 91

Code:

```
mse_multi = mean_squared_error(  
    y_test,  
    pred_multi  
)
```

```
mae_multi = mean_absolute_error(  
    y_test,  
    pred_multi  
)
```

```
rmse_multi = np.sqrt(mse_multi)
```

```
r2_multi = r2_score(  
    y_test,  
    pred_multi  
)
```

```
print("MSE :", mse_multi)
print("MAE :", mae_multi)
print("RMSE :", rmse_multi)
print("R2 Score :", r2_multi)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 92

Markdown Content:

This code evaluates the accuracy of the Multiple Linear Regression model using MSE, MAE, RMSE, and R^2 Score. These metrics compare the actual house prices with the predicted prices and help determine how well the model performs. Lower error values and a higher R^2 Score indicate better prediction accuracy.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 93

Markdown Content:

This code is used to evaluate the Multiple Linear Regression model. MSE, MAE, and RMSE measure the prediction errors, while the R^2 Score measures how well the model explains the variation in house prices. Lower error values and a higher R^2 Score indicate a more accurate model.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 94

Markdown Content:

MSE (Mean Squared Error) measures the average squared prediction error of the model. Lower MSE indicates better performance.

MAE (Mean Absolute Error) measures the average absolute difference between actual and predicted values. Lower MAE indicates higher accuracy.

RMSE (Root Mean Squared Error) measures the prediction error in the same unit as

the target variable. Lower RMSE indicates better predictions.

R² Score (Coefficient of Determination) measures how well the model explains the variation in the target variable. A value closer to 1 indicates a better model.

Adjusted R² Score (Adjusted Coefficient of Determination) measures model performance while considering the number of features used. A higher value indicates a better and more reliable model.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 95

Code:

```
plt.figure(figsize=(8,5))
```

```
plt.scatter(  
    y_test,  
    pred_multi,  
    alpha=0.5  
)
```

```
plt.xlabel("Actual Price")  
plt.ylabel("Predicted Price")
```

```
plt.title("Actual vs Predicted House Prices")
```

```
plt.show()
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 96

Markdown Content:

This code visulize actual VS pridictes House price .

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 97

Code:

```
residuals_multi = y_test - pred_multi
```

```
plt.figure(figsize=(8,5))
```

```
sns.scatterplot(  
    x=pred_multi,  
    y=residuals_multi  
)
```

```
plt.axhline(  
    y=0,  
    color='red'  
)
```

```
plt.xlabel("Predicted Values")  
plt.ylabel("Residuals")
```

```
plt.title("Residual Plot")
```

```
plt.show()
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 98

Markdown Content:

This code creates a Residual Plot for the Multiple Linear Regression model. The plot is used to check the prediction errors by comparing actual and predicted house prices. If the residuals are randomly scattered around the zero line, it indicates that the model is performing well and the assumptions of linear regression are satisfied.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 99

Code:

```
comparison = pd.DataFrame({
    'Model': [
        'Simple Linear Regression',
        'Multiple Linear Regression'
    ],
    'R2 Score': [
        r2_simple,
        r2_multi
    ]
})
```

comparison

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 100

Markdown Content:

This code used to compare Simple linear regression And mMultiple liner regression.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 101

Code:

```
plt.figure(figsize=(6,4))
```

```
sns.barplot(
    x='Model',
    y='R2 Score',
    data=comparison
)
```

```
plt.title("Simple vs Multiple Linear Regression")
```

```
plt.show()
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 102

Markdown Content:

This code visualize both regression in bar plot.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 103

Code:

```
from sklearn.preprocessing import PolynomialFeatures
```

```
poly2 = PolynomialFeatures(  
    degree=2,  
    include_bias=False  
)
```

```
X_train_poly2 = poly2.fit_transform(X_train)
```

```
X_test_poly2 = poly2.transform(X_test)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 104

Markdown Content:

This code creates polynomial features of degree 2 from the original dataset. The transformed features allow the Polynomial Regression model to capture non-linear patterns and improve prediction accuracy when the relationship between input features and house price is not strictly linear.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 105

Code:

```
poly_model2 = LinearRegression()
```

```
poly_model2.fit(  
    X_train_poly2,  
    y_train  
)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 106

Markdown Content:

This code creates and trains a Polynomial Regression model using the transformed polynomial features. The model learns the relationship between house characteristics and house prices and can later be used to make predictions on new data.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 107

Code:

```
pred_poly2 = poly_model2.predict(  
    X_test_poly2  
)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 108

Markdown Content:

This code generates house price predictions using the trained Polynomial Regression model. The predicted values are stored in `pred_poly2` and are used to evaluate the accuracy and performance of the model.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 109

Code:

```
mse_poly2 = mean_squared_error(  
    y_test,  
    pred_poly2  
)
```

```
mae_poly2 = mean_absolute_error(  
    y_test,  
    pred_poly2  
)
```

```
rmse_poly2 = np.sqrt(mse_poly2)
```

```
r2_poly2 = r2_score(  
    y_test,  
    pred_poly2  
)
```

```
print("MSE :", mse_poly2)  
print("MAE :", mae_poly2)  
print("RMSE :", rmse_poly2)  
print("R2 Score :", r2_poly2)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 110

Markdown Content:

This code calculates MSE (Mean Squared Error), MAE (Mean Absolute Error), RMSE (Root Mean Squared Error), and R^2 Score (Coefficient of Determination) to evaluate the accuracy of the Polynomial Regression model. Lower error values and a higher R^2 Score indicate better model performance and prediction accuracy.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 111

Code:

```
n = len(y_test)

p = X_test_poly2.shape[1]

adj_r2_poly2 = 1 - (
    ((1-r2_poly2)*(n-1))
    /
    (n-p-1)
)

print("Adjusted R2 :", adj_r2_poly2)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 112

Markdown Content:

This code calculates the Adjusted R^2 Score (Adjusted Coefficient of Determination) for the Polynomial Regression model. It measures how well the polynomial features explain the variation in house prices while considering the number of predictors used. A higher Adjusted R^2 value indicates better model performance.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 113

Code:

```
plt.figure(figsize=(8,5))
```

```
plt.scatter(  
    y_test,  
    pred_poly2,  
    alpha=0.5  
)
```

```
plt.xlabel("Actual Price")  
plt.ylabel("Predicted Price")
```

```
plt.title("Polynomial Regression Degree 2")
```

```
plt.show()
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 114

Markdown Content:

This code creates a scatter plot to compare the actual house prices with the predicted house prices generated by the Degree-2 Polynomial Regression model. If the predicted values are close to the actual values, the model is performing well.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 115

Code:

```
poly3 = PolynomialFeatures(  
    degree=3,  
    include_bias=False  
)
```

```
X_train_poly3 = poly3.fit_transform(  
    X_train  
)
```

```
X_test_poly3 = poly3.transform(  
    X_test  
)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 116

Markdown Content:

This code transforms the original dataset into Degree-3 polynomial features. These additional features help the Polynomial Regression model capture complex non-linear relationships and improve prediction accuracy when the relationship between input variables and house prices is not linear.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 117

Code:

```
poly_model3 = LinearRegression()  
  
poly_model3.fit(  
    X_train_poly3,  
    y_train  
)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 118

Markdown Content:

This code creates and trains a Degree-3 Polynomial Regression model using the transformed polynomial features. The model learns complex patterns in the data and can later be used to predict house prices more accurately.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 119

Code:

```
pred_poly3 = poly_model3.predict(  
    X_test_poly3  
)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 120

Markdown Content:

This code generates house price predictions using the trained Degree-3 Polynomial Regression model. The predicted values are stored in `pred_poly3` and are used to assess the accuracy and effectiveness of the model.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 121

Code:

```
mse_poly3 = mean_squared_error(  
    y_test,  
    pred_poly3  
)
```

```
mae_poly3 = mean_absolute_error(  
    y_test,  
    pred_poly3  
)
```

```
rmse_poly3 = np.sqrt(mse_poly3)
```

```
r2_poly3 = r2_score(  
    y_test,  
    pred_poly3  
)
```

```
print("MSE :", mse_poly3)  
print("MAE :", mae_poly3)  
print("RMSE :", rmse_poly3)  
print("R2 Score :", r2_poly3)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 122

Markdown Content:

This code calculates MSE (Mean Squared Error), MAE (Mean Absolute Error), RMSE (Root Mean Squared Error), and R^2 Score (Coefficient of Determination) to evaluate the accuracy of the Degree-3 Polynomial Regression model. Lower error values and a higher R^2 Score indicate better model performance and prediction accuracy.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 123

Code:

```
n = len(y_test)
```

```
p = X_test_poly3.shape[1]
```

```
adj_r2_poly3 = 1 - (  
    ((1-r2_poly3)*(n-1))  
    /  
    (n-p-1)  
)
```

```
print("Adjusted R2 :", adj_r2_poly3)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 124

Markdown Content:

This code calculates the Adjusted R^2 Score (Adjusted Coefficient of Determination) for the Degree-3 Polynomial Regression model. It measures how well the model explains the variation in house prices while considering the number of polynomial features used. A higher Adjusted R^2 value indicates better model performance.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 125

Code:

```
model_comparison = pd.DataFrame({
    'Model': [
        'Simple Linear Regression',
        'Multiple Linear Regression',
        'Polynomial Degree 2',
        'Polynomial Degree 3'
    ],
    'R2 Score': [
        r2_simple,
        r2_multi,
        r2_poly2,
        r2_poly3
    ]
})
```

model_comparison

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 126

Markdown Content:

This code calculates the Adjusted R^2 Score (Adjusted Coefficient of Determination) Compare all regression together.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 127

Code:

```
plt.figure(figsize=(8,5))
```

```
sns.barplot(  
    x='Model',  
    y='R2 Score',  
    data=model_comparison  
)
```

```
plt.title("Model Performance Comparison")
```

```
plt.xticks(rotation=15)
```

```
plt.show()
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 128

Markdown Content:

This code vizulize the Adjusted R^2 Score (Adjusted Coefficient of Determination) Compare all regression togetherusing barplot.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 129

Code:

```
best_model = model_comparison.loc[
    model_comparison['R2 Score'].idxmax()
]

print(best_model)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 130

Markdown Content:

This code identifies and displays the model with the highest R^2 Score from the model comparison table. Since a higher R^2 Score indicates better prediction performance

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 131

Code:

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(
    X_train
)

X_test_scaled = scaler.transform(
    X_test
)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 132

Markdown Content:

This code used to StandardScaler the data.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 133

Code:

```
from sklearn.linear_model import SGDRegressor
```

```
sgd = SGDRegressor(  
    max_iter=1000,  
    learning_rate='constant',  
    eta0=0.01,  
    random_state=42  
)
```

```
sgd.fit(  
    X_train_scaled,  
    y_train  
)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 134

Markdown Content:

This code trains an SGD Regressor model using scaled training data to predict house prices and compare its performance with other regression algorithms

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 135

Markdown Content:

SGD Regressor = Stochastic Gradient Descent Regressor

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 136

Code:

```
pred_sgd = sgd.predict(  
    X_test_scaled  
)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 137

Markdown Content:

This code generates house price predictions using the trained SGD regression model.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 138

Code:

```
mse_sgd = mean_squared_error(  
    y_test,  
    pred_sgd  
)
```

```
mae_sgd = mean_absolute_error(  
    y_test,  
    pred_sgd  
)
```

```
rmse_sgd = np.sqrt(mse_sgd)
```

```
r2_sgd = r2_score(  
    y_test,  
    pred_sgd  
)  
  
print("MSE :", mse_sgd)  
print("MAE :", mae_sgd)  
print("RMSE :", rmse_sgd)  
print("R2 Score :", r2_sgd)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 139

Markdown Content:

This code calculates MSE (Mean Squared Error), MAE (Mean Absolute Error), RMSE (Root Mean Squared Error), and R^2 Score (Coefficient of Determination) to evaluate the accuracy of the Degree-3 Polynomial Regression model. Lower error values and a higher R^2 Score indicate better model performance and prediction accuracy.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 140

Code:

```
train_pred_simple = slr.predict(X_train_s)  
  
train_r2_simple = r2_score(  
    y_train_s,  
    train_pred_simple  
)  
  
print("Training R2 :", train_r2_simple)  
print("Testing R2 :", r2_simple)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 141

Markdown Content:

It tells how well your model is performing on both trained data and new data.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 142

Code:

```
train_pred_multi = mlr.predict(X_train)
```

```
train_r2_multi = r2_score(  
    y_train,  
    train_pred_multi  
)
```

```
print("Training R2 :", train_r2_multi)  
print("Testing R2 :", r2_multi)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 143

Markdown Content:

This code It tells how well your model is performing on both trained data and new data.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 144

Code:

```
train_pred_poly2 = poly_model2.predict(  
    X_train_poly2  
)
```

```
train_r2_poly2 = r2_score(  
    y_train,  
    train_pred_poly2  
)
```

```
print("Training R2 :", train_r2_poly2)  
print("Testing R2 :", r2_poly2)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 145

Markdown Content:

Compare difference between before and after the accuracy .

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 146

Code:

```
train_pred_poly3 = poly_model3.predict(  
    X_train_poly3  
)
```

```
train_r2_poly3 = r2_score(  
    y_train,  
    train_pred_poly3  
)
```

```
print("Training R2 :", train_r2_poly3)  
print("Testing R2 :", r2_poly3)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 147

Markdown Content:

Compare difference between before and after the accuracy .

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 148

Code:

```
bias_variance = pd.DataFrame({
    'Model': [
        'Simple Linear',
        'Multiple Linear',
        'Polynomial Degree 2',
        'Polynomial Degree 3'
    ],
    'Training R2': [
        train_r2_simple,
        train_r2_multi,
        train_r2_poly2,
        train_r2_poly3
    ],
    'Testing R2': [
        r2_simple,
        r2_multi,
        r2_poly2,
        r2_poly3
    ]
})

bias_variance
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 149

Markdown Content:

Compare the model together

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 150

Code:

```
bias_variance.set_index('Model').plot(  
    kind='bar',  
    figsize=(10,5)  
)
```

```
plt.title("Training vs Testing R2")
```

```
plt.ylabel("R2 Score")
```

```
plt.show()
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 151

Markdown Content:

This code used to visulize Training vs Testing R2

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 152

Code:

```
bias_variance['Difference'] = (  
    bias_variance['Training R2']  
    -  
    bias_variance['Testing R2']  
)
```

bias_variance

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 153

Markdown Content:

This code uaed to compare all regression together .

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 154

Code:

```
model_comparison.sort_values(  
    by='R2 Score',  
    ascending=False  
)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 155

Markdown Content:

This code used to compare rate of the R2

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 156

Code:

```
final_results = pd.DataFrame({
    'Model':[
        'Simple Linear Regression',
        'Multiple Linear Regression',
        'Polynomial Degree 2',
        'Polynomial Degree 3',
        'SGD Regressor'
    ],

    'R2 Score':[
        r2_simple,
        r2_multi,
        r2_poly2,
        r2_poly3,
        r2_sgd
    ],

    'RMSE':[
        rmse_simple,
        rmse_multi,
        rmse_poly2,
        rmse_poly3,
        rmse_sgd
    ]
})
```

final_results

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 157

Markdown Content:

This code used to compare the Regression .

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 158

Code:

```
best_model = final_results.loc[
    final_results['R2 Score'].idxmax()
]
```

```
print(best_model)
```

Purpose: This code cell performs the operation defined in the notebook workflow. It contributes to data loading, preprocessing, visualization, model building, evaluation, or comparison depending on its position in the pipeline.

Working: The code is executed sequentially as part of the machine learning workflow. Variable names, functions, and transformations are preserved exactly as implemented in the notebook.

STEP 159

Markdown Content:

This code selects the most accurate model from all tested models and displays it.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 160

Markdown Content:

The main objective of this project was to predict house prices using machine learning techniques. The dataset contained important property features such as House Area, Number of Bedrooms, Number of Bathrooms, Location Score, and Age of Property. Before building the models, data preprocessing steps such as checking missing values, handling outliers, and performing exploratory data analysis were carried out to improve data quality and gain insights into the dataset.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 161

Markdown Content:

Different regression models were implemented, including Simple Linear Regression, Multiple Linear Regression, Polynomial Regression (Degree 2 and Degree 3), and SGD Regression. The performance of each model was evaluated using MSE, MAE, RMSE, R^2 Score, and Adjusted R^2 Score.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 162

Markdown Content:

The results showed that Simple Linear Regression provided the lowest accuracy because it used only one feature for prediction. Multiple Linear Regression performed better by considering all important features. Polynomial Regression achieved the best performance by capturing both linear and non-linear relationships between property characteristics and house prices. Therefore, it was identified as the most effective model for this dataset.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

Conclusion

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

STEP 164

Markdown Content:

In conclusion, the project successfully achieved its goal of predicting house prices and comparing different regression models. Among all the models tested, Polynomial Regression provided the most accurate predictions and was selected as the best-performing model for house price prediction.

Interpretation: This markdown cell documents the reasoning, explanation, or observations associated with the surrounding code cells.

Project Summary

The notebook implements an end-to-end house price prediction workflow including dataset loading, EDA, missing-value analysis, outlier detection, feature selection, train-test splitting, Simple Linear Regression, Multiple Linear Regression, Polynomial Regression (Degree 2 and Degree 3), SGD Regression, model evaluation using MSE,

MAE, RMSE, R^2 and Adjusted R^2 , residual analysis, bias-variance comparison, and final model selection.